Ahmedabad University

Subject - Digital Image Processing

8. Vehicle Orientation Estimation Using Conventional Image Processing

Weekly Report - 6

Date of Submission: 31/10/2026

Repo Link: https://github.com/encrypted-official/DIP_2026_Project8_Group1

Group - 1

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2026 (Monsoon Semester)

1. Work done this week

Last week we ran everything on the first 200 validation images and left the bigger runs for this week. The plan was to go to 500 images first and then to the full split. Since the scripts were already stable, we skipped the 500 image step and ran directly on the whole validation split, using an image limit of 1,200, which covers all 1,184 validation images. Everything in this report comes from that run (the week6_1200 output folder). We also added the shadow check for Method C, reported Method C on a fair common set of crops, and looked at why the methods fail where they do. The front vs back step did not start this week and moves to next week.

Full validation run: Cutting crops from the full split gave 52,576 crops, which is about 5.8 times the 9,086 crops we used last week. Looking at the labels of these crops, the dataset is strongly tilted towards vehicles lying along the image axes. 80.7% of the vehicles have a true axis within 15 degrees of horizontal or vertical, 19.4% of the labels are exactly 0, 90 or 180 degrees, and 48.1% are whole numbers. This matters later in the report, because a method that is good on axis aligned vehicles will look good on the overall average even if it fails on the rest.

We first ran all three methods with the same default settings as Table 2 of last week (blur 3, CLAHE on, Canny 50 and 150) so that the two weeks can be compared directly. Table 1 shows the result.

Table 1: Default settings on 200 images (Week 5) against the full validation split (Week 6)

Metric	PCA (200 img)	PCA (full)	Hough (200 img)	Hough (full)	Otsu + mom. (200 img)	Otsu + mom. (full)
Crops	9,086	52,576	9,086	52,576	9,086	52,576
Coverage (%)	99.88	99.95	99.80	99.95	94.78	99.02
MAE (°)	9.46	7.84	12.31	8.86	10.74	11.68
Median error (°)	5.58	4.25	6.12	4.56	4.87	6.37
P90 error (°)	23.91	17.87	33.47	18.96	27.86	30.24
Within 10° (%)	69.83	77.53	66.12	76.15	71.04	64.34

The two edge methods got clearly better on the full split. PCA dropped from 9.46 to 7.84 degrees MAE and Hough from 12.31 to 8.86, with the P90 of Hough falling by almost 15 degrees. Method C went the other way. Its coverage rose from 94.78% to 99.02%, but its MAE went up by about a degree and its median went from the best of the three to the worst. So the first 200 images were not a fair sample of the split. They seem to be harder for the edge methods and easier for the blob method than the rest of the data. Last week we wrote that the 200 image

numbers were close to the 40 image re-run and so "not a lucky picture". That was true between those two sets, but both came from the start of the split, and the full run shows that conclusions drawn on a slice of the data can change.

Ablation on the full split: We ran ablation.py over the full 52,576 crops with the same grid as last week, plus the new shadow check switch for Method C. That gives 18 settings each for PCA and Hough and 12 for Method C (6 blur and CLAHE settings, each with the shadow check on and off), so 48 runs in total. Before the big run, we tested the updated script on the first 100 crops only (the ablation_test folder). On those 100 crops the best setting was Hough at blur 1, CLAHE on and (50, 150) with an MAE of 5.70 degrees. On the full split the same setting gives 7.79. This was a useful warning about trusting a ranking from a small set. Figure 1 shows MAE against blur size on the full split, in the same layout as last week.

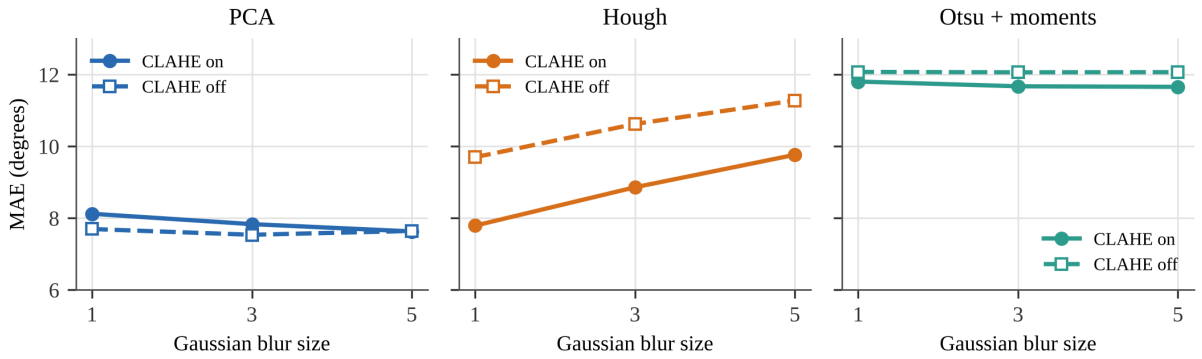


Figure 1: MAE against blur size with CLAHE on and off, full validation split (Canny 50, 150 for PCA and Hough; Method C without the shadow check)

Several of last week's patterns did not hold on the full split. For Hough, more blur now makes things worse, not better. MAE goes from 7.79 at blur 1 to 9.77 at blur 5 with CLAHE on. CLAHE helps Hough at every setting, by 0.8 to 1.9 degrees. For PCA the lines are almost flat and CLAHE off is slightly better. For Method C, CLAHE now helps by 0.3 to 0.4 degrees, which is the opposite of what we saw on 200 images. Figure 2 shows the effect of the Canny thresholds at blur 3.

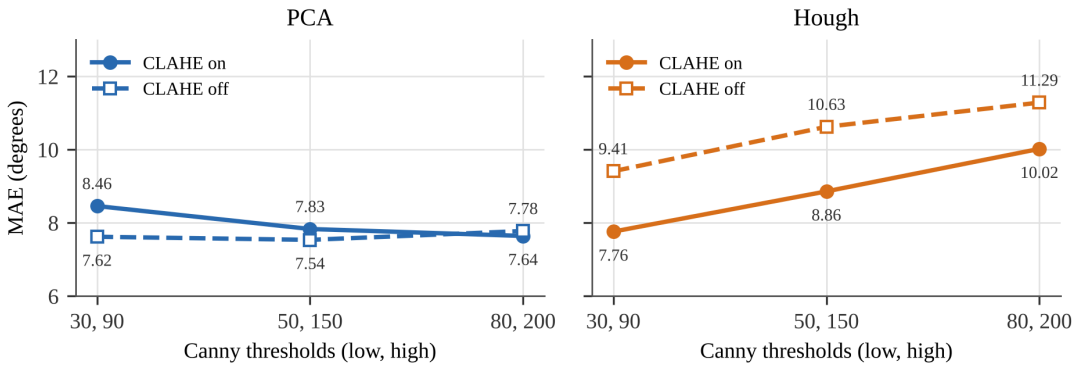


Figure 2: MAE against Canny thresholds at blur 3, full validation split

The Canny result is also reversed for Hough. Last week the higher pair (80, 200) helped. On the full split the lower pair (30, 90) is best, and moving up to (80, 200) adds more than 2 degrees. Our reading is that on the full split most crops are clean, so the extra edges from a lower threshold are mostly real body edges. More edges give HoughLinesP more points to vote with, and the long side of the car gets found more often. PCA is much less sensitive to the thresholds because it uses all edge points at once, so a few extra or missing points do not move the axis much. Table 2 compares the setting we picked last week with the best setting on the full split.

Table 2: Week 5 pick against the best setting on the full split (52,576 crops)

Method	Setting	Blur	CLAHE	Canny	Cov. (%)	MAE (°)	Median (°)	P90 (°)	≤10° (%)
PCA	Week 5 pick	5	On	80, 200	99.94	7.68	4.34	17.11	78.26
	Full-split best	5	Off	30, 90	99.96	7.42	4.05	16.76	79.01
Hough	Week 5 pick	3	On	80, 200	99.92	10.02	4.92	22.04	73.60
	Full-split best	1	On	30, 90	99.98	7.58	4.25	17.83	78.61
Otsu + moments	Week 5 pick	5	Off	N/A	98.69	12.07	7.05	30.60	62.48
	Full-split best	5	On	N/A	99.02	11.66	6.33	30.25	64.25

For PCA, last week's pick is still close to the best (0.25 degrees behind). For Hough, it is now one of the worse settings in the grid, 2.4 degrees behind the best and 1.2 degrees worse than the plain default. With the full-split best settings, PCA and Hough end up only 0.15 degrees apart in MAE, so the clear gap we reported in earlier weeks mostly came from tuning Hough on too small a set. Unlike last week, all three best settings keep coverage at or above 99%, so there was no coverage trade-off to make this time. We will use the full-split settings from now on.

Fair comparison for Method C: As planned, we report Method C both on its own valid crops and on the crops that all three methods answered. This uses the prediction files from the comparison run, which used the defaults for PCA and Hough and last week's pick (blur 5, CLAHE off) for Method C. Table 3 shows both views.

Table 3: Each method on its own valid crops and on the 51,839 crops all three answered

Method	Crops used	Valid	Cov. (%)	MAE (°)	Median (°)	P90 (°)	≤10° (%)
PCA	Own valid	52,550	99.95	7.84	4.25	17.87	77.53
Hough	Own valid	52,552	99.95	8.86	4.56	18.96	76.15
Otsu + moments	Own valid	51,888	98.69	12.07	7.05	30.60	62.48
PCA	Common	51,839	100.00	7.71	4.23	17.52	77.84
Hough	Common	51,839	100.00	8.85	4.55	18.95	76.29
Otsu + moments	Common	51,839	100.00	12.05	7.04	30.53	62.51

On the full split the question of fairness turned out to be small, because Method C now skips only 688 crops (1.31%). Those 688 crops are hard ones, as we expected. PCA has an MAE of 16.91 degrees on them against 7.71 on the rest. Removing them only moves PCA by 0.12 degrees and Method C by 0.02 degrees, so the ranking is the same either way. Figure 3 shows the full error curve on the common crops.

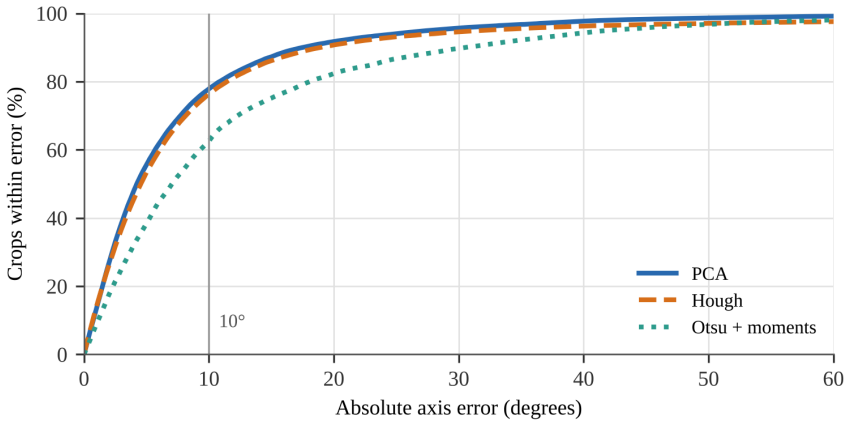


Figure 3: Share of crops within a given error, on the 51,839 common crops

PCA and Hough are almost on top of each other, with PCA a little ahead everywhere. Method C is lower across the whole curve. It has only 38.3% of crops within 5 degrees, against 55.6% for

PCA and 53.4% for Hough. So last week's finding that Method C has the lowest median does not hold on the full split.

Shadow check for Method C: Following last week's idea, the new check in estimate_moments.py looks at the chosen blob and, when its darkest part sits on one side of the blob only, removes that part before computing the moments. Table 4 shows the result at blur 5 with CLAHE off.

Table 4: Method C with and without the shadow check (blur 5, CLAHE off)

Metric	Without check	With check
Valid / total	51,888 / 52,576	52,202 / 52,576
Coverage (%)	98.69	99.29
MAE (°)	12.07	12.23
Median error (°)	7.05	6.92
P90 error (°)	30.60	30.82
Within 10° (%)	62.48	63.58
MAE on 51,733 crops both answered (°)	11.99	12.05

The shadow check does not help. It changes the prediction by more than 2 degrees on 40.4% of the crops, but among those it improves the error only 49.0% of the time, which is the same as a coin toss. It also lets Method C answer 469 crops that used to be skipped, with an MAE of 32.26 degrees on them, which is why the overall MAE goes up even though the median goes down a little. In all six blur and CLAHE settings, turning the shadow check on made the MAE worse, by 0.06 to 0.37 degrees. We think the rule is too simple. A dark region on one side of the blob is just as often a dark windscreen or roof part as it is a shadow, and our check cannot tell them apart without colour or sun direction information.

Error by vehicle orientation: Since we did not have crop sizes in this week's per-crop file, we split the errors by the true angle instead. Figure 4 shows the MAE in 15 degree bins, with the number of crops in each bin below it.

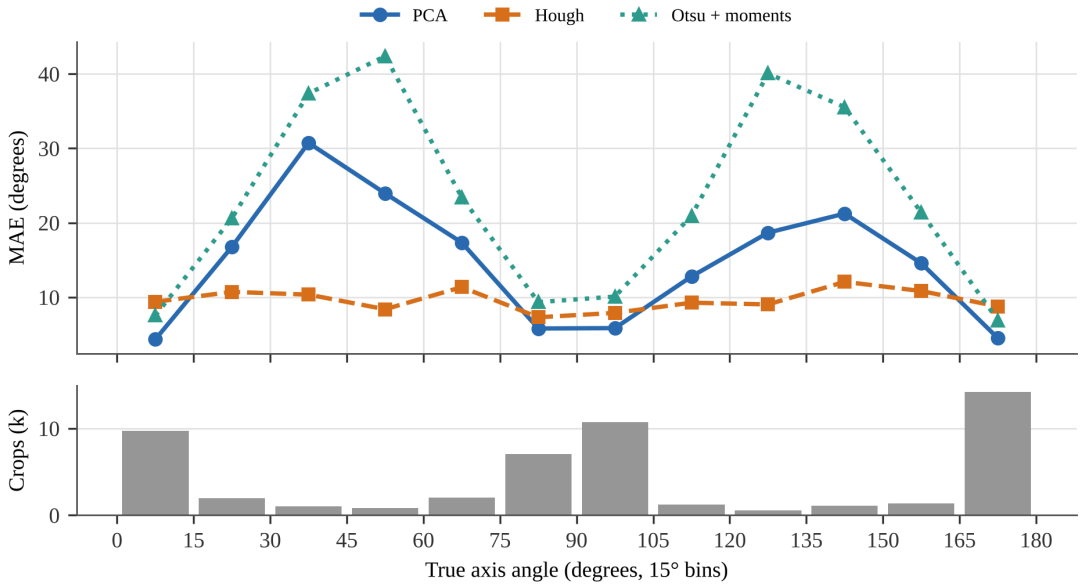


Figure 4: MAE by true axis angle on the 51,839 common crops (top) and crops per bin (bottom)

This is the most useful result of the week. PCA and Method C are very good when the vehicle lies along an image axis, but their error rises sharply for diagonal vehicles, reaching 31 degrees (PCA) and 42 degrees (Method C) for vehicles between 30 and 60 degrees. Hough stays between about 7 and 12 degrees at every angle. Table 5 splits the common crops into axis aligned vehicles (within 15 degrees of horizontal or vertical) and diagonal vehicles.

Table 5: MAE, median and within 10° for axis aligned and diagonal vehicles (common crops)

Method	Axis aligned (41,811 crops, 80.7%)			Diagonal (10,028 crops, 19.3%)		
	MAE (°)	Median (°)	≤10° (%)	MAE (°)	Median (°)	≤10° (%)
PCA	5.06	3.48	86.9	18.76	13.47	40.1
Hough	8.46	4.30	78.7	10.46	5.97	66.1
Otsu + moments	8.31	5.62	72.8	27.65	25.79	19.6

On diagonal vehicles Hough is far better than PCA (10.46 against 18.76 degrees). PCA only wins overall because four out of five vehicles in this dataset are axis aligned. We also checked the direction of the errors on the diagonal vehicles. The PCA prediction is closer to the nearest image axis than the true angle in 82.7% of the cases, by 9.7 degrees on average, and Method C does the same in 86.8% of the cases, by 13.8 degrees. Hough shows almost no such pull (53.2%, 1.8 degrees). Our explanation is that the crops are axis aligned boxes. For a diagonal vehicle the box

also contains road corners, lane marks and parts of neighbouring vehicles, and the box itself is wider in one image direction. PCA and the Otsu blob use every edge point or pixel in the box, so this background pulls the axis towards the box sides. Hough only keeps long straight lines, and the long sides of the car usually win the vote.

Combining PCA and Hough: Since the two methods fail in different places, we tried some simple combinations on the same predictions. Table 6 shows the results on the common crops.

Table 6: Simple combinations of the existing predictions (51,839 common crops)

Approach	MAE (°)	Median (°)	P90 (°)	≤10° (%)
PCA alone	7.71	4.23	17.52	77.84
Hough alone	8.85	4.55	18.95	76.29
Axial mean of PCA and Hough	7.35	4.17	17.22	79.29
PCA if its answer is within 15° of an axis, else Hough	7.06	4.07	16.46	79.18
Axial mean of all three	6.73	4.23	15.87	78.40

All three combinations beat the best single method. The switch rule uses only the PCA answer to decide, so it can be applied at test time without the truth. It cuts the diagonal MAE from 18.76 to 15.51 degrees while keeping the axis aligned MAE at 5.03. The axial mean of all three gives the lowest MAE, mainly by pulling in the worst cases. One caveat is that we chose the 15 degree switch point on the same crops we evaluated on, so these numbers are a little optimistic until we check them on the train split. Agreement between PCA and Hough also works as a confidence signal. When the two are within 5 degrees of each other (50.6% of crops), PCA has an MAE of only 4.30 degrees. When they disagree by more, its MAE is 11.20.

2. Work to be done next week

Week 7 in our plan is the small crops, and the front vs back step moves there from this week. We plan to:

- Add the crop width and height to the per-crop output, so that the error by crop size from last week's Figure 2 can be redone on the full split.
- Re-run the three methods with the full-split settings from Table 2, and check the combination rules from Table 6 on the train split, so that the switch point is not chosen on the same data it is tested on.
- Reduce the background pull on diagonal vehicles, for example by keeping only edge points inside an ellipse around the crop centre, or by using the Hough answer to rotate the crop before running PCA.
- Drop the current shadow check, or replace it with one that uses colour (shadows keep their hue while getting darker), and only keep it if it helps on the common crops.
- Start front vs back by looking at the intensity profile along the predicted axis, since the windscreen and the rear window usually look different.

3. Literature Review

This week was mostly about Method C and the failures on diagonal vehicles, so we read about thresholding, moments, shadows and the problem with axis aligned boxes.

- Otsu (1979) chooses the threshold that maximises the between-class variance of the grey level histogram. This assumes two clear peaks, which explains why Method C has nothing to split on grey cars on grey road.
- Sezgin and Sankur (2004) compare a large number of thresholding methods and show that no single one is best across image types. Local or entropy based thresholds could be worth trying for the low contrast cases.
- Hu (1962) introduced image moments for shape description. The orientation from second order central moments is the angle of the best fit ellipse, so anything that joins the blob, such as a shadow or a background patch, shifts the axis directly.
- Prati et al. (2003) and Finlayson et al. (2006) describe shadow detection. Both use the fact that a shadow mostly lowers brightness but keeps the colour of the surface. Our grey level shadow check does not use this, which may be why it could not tell shadows from dark car parts.

- Xia et al. (2018), who built the DOTA aerial dataset, point out that horizontal boxes around rotated objects contain a lot of background. This is the same effect we found in Figure 4 for diagonal vehicles inside axis aligned crops.

4. Challenges Faced

- Settings that do not carry over. The settings we picked on 200 images were wrong for Hough on the full split, and the 100 crop test was even more misleading. From now on we will only pick settings on the full split, or on the train split when we can.
- Diagonal vehicles. They are only about a fifth of the data, but PCA and Method C have errors two to three times higher on them, and the overall MAE hides this.
- The shadow check. It did not work, and separating a shadow from a dark part of the car with grey levels alone seems harder than we thought.
- Missing crop sizes. The per-crop file of this run did not keep the crop dimensions, so we could not repeat the size breakdown on the full split this week.

References

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[3] M.-K. Hu, "Visual Pattern Recognition by Moment Invariants," IRE Transactions on Information Theory, vol. 8, no. 2, pp. 179-187, 1962.

[4] A. Prati, I. Mikic, M. M. Trivedi and R. Cucchiara, "Detecting Moving Shadows: Algorithms and Evaluation," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 25, no. 7, pp. 918-923, 2003.

[5] G. D. Finlayson, S. D. Hordley, C. Lu and M. S. Drew, "On the Removal of Shadows from Images," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 28, no. 1, pp. 59-68, 2006.

[6] G.-S. Xia et al., "DOTA: A Large-Scale Dataset for Object Detection in Aerial Images," in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 3974-3983.

[7] DRASHTI-HaOBB dataset, Zenodo, https://zenodo.org/records/18278989